MIL-C-60220(MU)
1 March 1965
SUPERSEDING
FA(FC)-PD-173
14 August 1962

MILITARY SPECIFICATION

CABLES, SPECIAL PURPOSE, ELECTRICAL
(DATA TRANSMISSION USE)

1. SCOPE

1.1 This specification covers flexible multiconductor cables, portable, for interconnecting various units of complex weapon systems where voltages do not exceed 600 volts root mean square (rms) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

J-C-98	Cable and Wire, Insulated Methods of Sampling and Testing
L-P-390	Plastic Molding Material, Polyethylene, Low and Medium Density

Military

MIL-C-17	Cables, Radio Frequency; Coaxial, Dual Coaxial, Twin Conductor and Twin Lead
MIL-C-13777	Cable, Special Purpose, Electrical: General Specification for

STANDARDS

Military

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6145

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3. REQUIREMENTS

3.1 Requirements for individual cable types.- The cable shall meet the requirements of MIL-C-13777 with the exceptions as specified herein and on Figures 1 to 3 inclusive for particular types of cables (see 6.2).

3.2 Conductor.- Copper stranded conductors shall be constructed as specified on the applicable figure. Steel reinforcing strands shall not be used.

3.3 Insulated compound.- There shall be applied over the conductor a well centered virgin insulating material consisting of extruded fluorinated ethylene propylene (F.E.P.).

3.3.1 Physical properties.

3.3.1.1 Ultimate elongation and tensile strength.- Insulation compound F.E.P. shall meet the following:

Ultimate elongation (min.) inches, 2-5
Tensile strength (min.) p.s.i., 2000

3.3.1.2 Shrinkage.- The shrinkback of insulation from either end of the conductor shall not exceed 1/16 inch and there shall be no evidence of ends splitting or cracking.

3.3.1.3 Electrical properties.- Insulation resistance and water absorption shall be as specified for Insulation Comp F in MIL-C-13777.

3.3.1.4 Voltage breakdown (primary insulation).- Prior to cabling, all insulated conductors shall show no evidence of breakdown when subjected to the voltage specified below:

AWG	VOLTS
20	1000

3.4 Filler.- Filler material shall be one of the following materials and shall meet all performance requirements specified herein.

Clean, dry, long fiber cotton, fungus treated; oriented polyethylene multi-monofilament filler; oriented polypropylene multi-monofilament filler; or twisted polyethylene film conforming to L-P-390, Type I, Grade 2 or 3. Where round fillers such as cores or wire replacements are required polyethylene rod filament conforming to L-P-390, Type II, Grade 7 may be used.

3.5 Cold bend torque.- The amount of torque required to bend the cable shall not exceed 30 foot pounds.

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3.6 Impact, bend and twist requirements.- These requirements are not applicable to cable specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The inspection requirements of MIL-C-13777 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.- The characteristics listed below are specified in 3.2 to 3.6 inclusive as requirements that are in exception of those specified in Section 3 of MIL-C-13777 (see 3.1) and shall be assigned to the respective Groups under "Acceptance inspection" in MIL-C-13777 as indicated for lot size and sampling:

	CHARACTERISTIC	REQUIREMENT	INSPECTION GROUP (MIL-C-13777)
101.	Conductor	3.2	A
102.	Insulated compound	3.3	B
103.	Elongation and tensile	3.3.1.1	B
104.	Shrinkage	3.3.1.2	B
105.	Electrical properties	3.3.1.3	A
106.	Voltage breakdown	3.3.1.4	A
107.	Filler	3.4	A
108.	Cold bend torque	3.5	B

4.3 Test methods and procedures.

4.3.1 Tests for individual cable types.- Except where otherwise identified below, tests to determine compliance with Section 3 shall be conducted in accordance with the applicable tests defined in MIL-C-13777.

4.3.2 Insulation compound tests.- The insulating compound shall be tested in accordance with the following physical and electrical tests to determine compliance with 3.3.

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4.3.2.1 Aging.- Insulation compound F.E.P. shall be oven aged for 96 hours at a temperature of 478 to 482°F. After oven aging the compound shall be tested in accordance with 4.3.2.2.

4.3.2.2 Shrinkage and dielectric test.- Remove 1-inch of insulation from each end of a 24-inch test specimen. Hang loose coil in an oven in accordance with 4.3.2.1. After exposure return temperature to room temperature, measure shrinkage within 1 hour for conformance with 3.3.1.2. Following oven aging specimens shall be wound tightly around a mandrel not less than three times the diameter of the finished wire for five close turns and removed from the mandrel as a helical coil. No cracks or splits shall be visible. The coiled specimens shall be immersed (except for approximately 2 inches at each end) in tap water at room temperature, for one hour and shall withstand for one minute the dielectric test voltage as covered in 3.3.1.4.

4.3.3 Voltage breakdown (insulated conductors).- The insulated conductors shall be tested 100 percent by the cable manufacturer in accordance with Method 6111 of J-C-98 except that the soak time shall be reduced to 4 hours without stirring the bath and with both ends of the conductor tied together and connected to the high voltage. Braided conductors shall be tested before applying the braid. Insulated conductors, of shielded wires shall be tested dry against the shield. There shall be no evidence of breakdown when tested at the voltages of 3.3.1.4.

4.4 Filler.- Data shall be supplied with each order certifying that when cotton is used as a filler material (3.4) the cotton has been fungus treated and conforms to the fungus test in accordance with Method 508 of MIL-STD-810.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with that specified in MIL-C-13777.

6. NOTES

6.1 Intended use.- The cable covered by this specification is intended for electrical and electronic data applications in protected areas where severe mechanical abuse is not inherent such as racks and trays, tunnels, wire ways, fastened on carriage components or drawn through metal conduits.

6.1.1 The design configuration for the cable is depicted on the applicable Figure as follows:

Figure 1 Data transmission cable for Reproducer, Signal Data AN/GSQ-64.

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Figure 2 Data transmission cable for Control Box, Computer Logic Unit, Test Set, C-4020/GSM-70.

Figure 3 Data transmission cable for Test Set, Computer Logic Unit, AN/GSM-70.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and length of cable required. (See 6.1.1).
(c) Selection of an applicable level of preservation, packaging and packing.

6.3 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon specific request of the contractor.

Custodian:

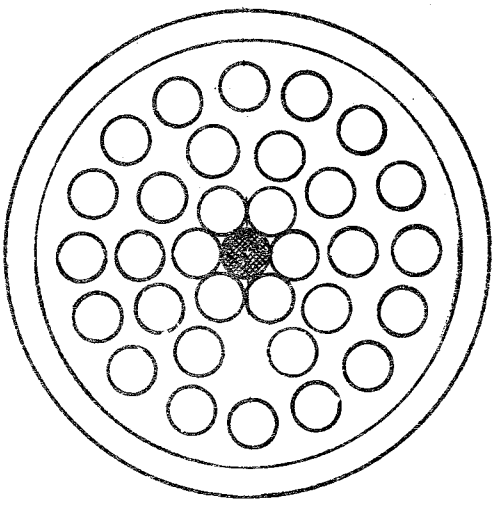
Army - MU

Preparing activity:

Army - MU

Project No. 6145-A050

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1. Total Wires		32 As Shown
2. Conductors		
A. 32 Single Conductors		
(1) Conductor	No. 20 AWG (19-32 Tinned Copper Strands)	
(2) Insulation		
(a) Min Thickness Comp. FEP		.006
(b) Total Diameter		.057 ± .002
(c) Color		White
(d) Spark Test Voltage		2000
(e) Inspection Test Voltage		1000
3. Cabling		Filler
A. Layer No. 1		
B. Layer No. 2		6
(1) No. of Wires		
C. Layer No. 3		10
(1) No. of Wires		
D. Layer No. 4		16
(1) No. of Wires		

DATA TRANSMISSION CABLE FOR REPRODUCER, SIGNAL DATA AN/GSQ-64

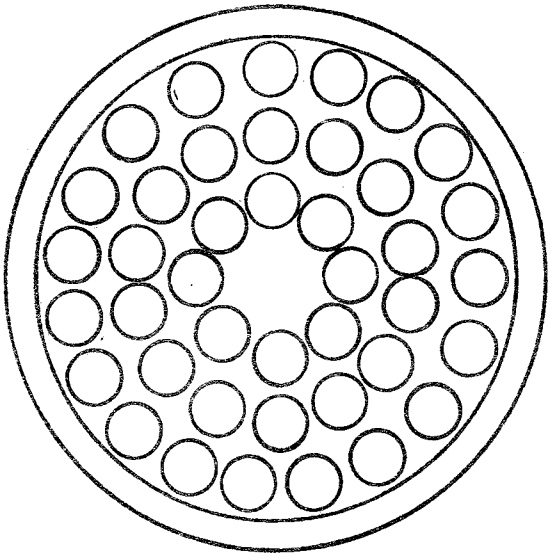
FIGURE 1

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FIGURE 1 (Cont'd)

4. Filler	Polyamide
5. Cable Separator	Polyethylene Terephalate
6. Barrier Tape	
7. Sheath per MIL-C-13777	
A. No. of Layers	1
B. Total Thickness Min.	.050
8. Cable	
A. Min. O.D.	.540
B. Max. O.D.	.580
C. Max. Weight (lb/1000 ft.)	500

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1. Total Wires		41 As Shown
2. Conductors		
A. 41 Single Conductors		
(1) Conductor	No. 20 AWG (19-32 Tinned Copper Strands)	
(2) Insulation		
(a) Min Thickness Comp. FEP		.006
(b) Total Diameter		.057 ± .002
(c) Color		White
(d) Spark Test Voltage		2000
(e) Inspection Test Voltage		1000
3. Cabling		
A. Layer No. 1		
(1) No. of Wires		8
B. Layer No. 2		
(1) No. of Wires		14
C. Layer No. 3		
(1) No. of Wires		19

DATA TRANSMISSION CABLE FOR CONTROL BOX, COMPUTER LOGIC UNIT, TEST SET C-4020/GSM-70

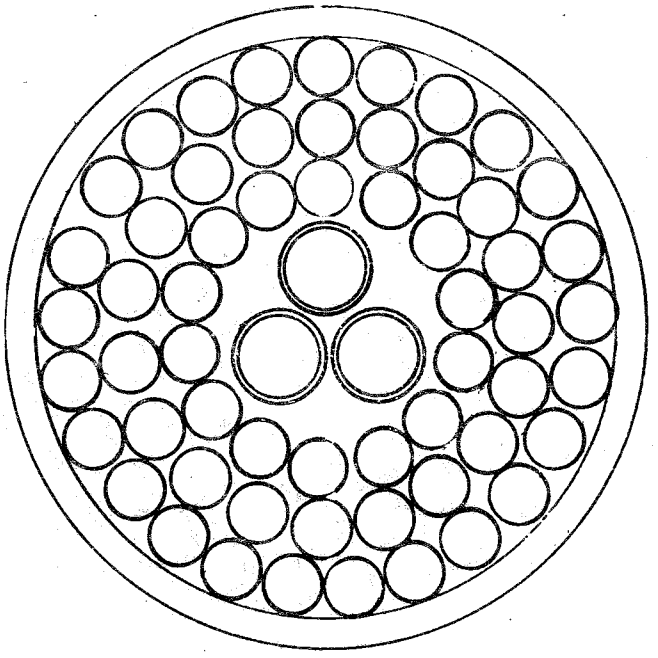
FIGURE 2

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FIGURE 2 (Cont'd)

4. Filler	Polyamide
5. Cable Separator	Polyethylene Terephalate
6. Barrier Tape	
7. Sheath per MIL-C-13777	
A. No. of Layers	1
B. Total Thickness Min.	.050
8. Cable	
A. Min. O.D.	.540
B. Max. O.D.	.580
C. Max. Weight (lb/1000 ft.)	500

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1. Total Wires	61 As Shown
2. Conductors	
A. Three Coaxials:	
(1) Inner Conductor:	No. 27 AWG stranded, 7 strands each of No. 35 AWG annealed (copper clad) steel wire per MIL-C-17. Nominal strand diameter 0.0056 in.
(2) Core:	Solid Type A (Polyethylene) per MIL-C-17. Diameter 0.097 ± 0.004 in.
(3) Outer Conductor:	Single braid, No. 38 AWG tinned copper wire, maximum diameter 0.118 in.
(4) Over Braid:	Braid, colored one each black, white and red lacquered. Maximum diameter 0.136 in.

DATA TRANSMISSION CABLE FOR TEST SET,
COMPUTER LOGIC UNIT, AN/GSM-70

FIGURE 3

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FIGURE 3 (Cont'd)

(5) Electrical Tests:	
Capacitance	20.5 ± 2 pf/ft
Attenuation (Max.)	13db/100 ft @ 400 mc
Impedance	73 ± 5 Ohms
B. 58 Single Conductors	
(1) Conductor	No. 20 AWG (19-#32 Tinned Copper Strands)
(2) Insulation	
(a) Min Thickness Comp. FEP	.006
(b) Total Diameter	.057 ± .002
(c) Color	White
(d) Spark Test Voltage	2000
(e) Inspection Test Voltage	1000
3. Cabling	
A. Layer No. 1	
(1) No. of Wires	3 Coaxial Sh.
(2) AWG No.	27
B. Layer No. 2	
(1) No. of Wires	14
(2) AWG No.	20
C. Layer No. 3	
(1) No. of Wires	19
(2) AWG No.	20
D. Layer No. 4	
(1) No. of Wires	25
(2) AWG No.	20
E. Maximum Lay	10"
4. Filler	Polyamide
5. Cable Separator	Polyethylene Terephalate
6. Barrier Tape	
7. Sheath per MIL-C-13777	
A. No. of Layers	1
B. Total Thickness Min.	.060
8. Cable	
A. Min. O.D.	.750
B. Max. O.D.	.790
C. Max. Weight (lb/1000 ft.)	510

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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